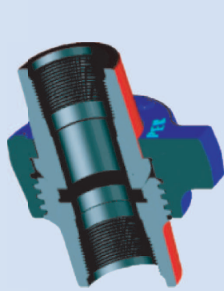
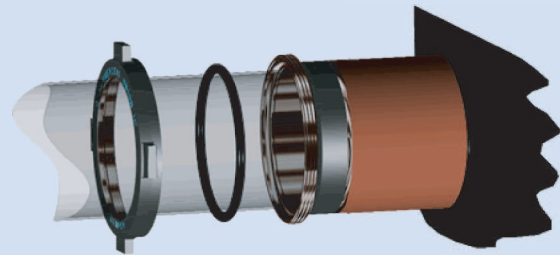


# KEMPER

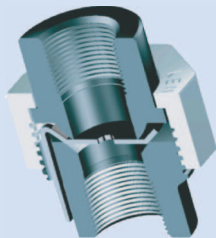
VALVE & FITTINGS CORPORATION



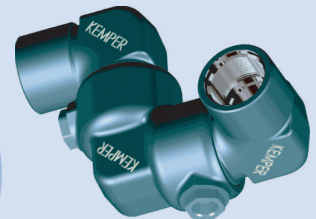
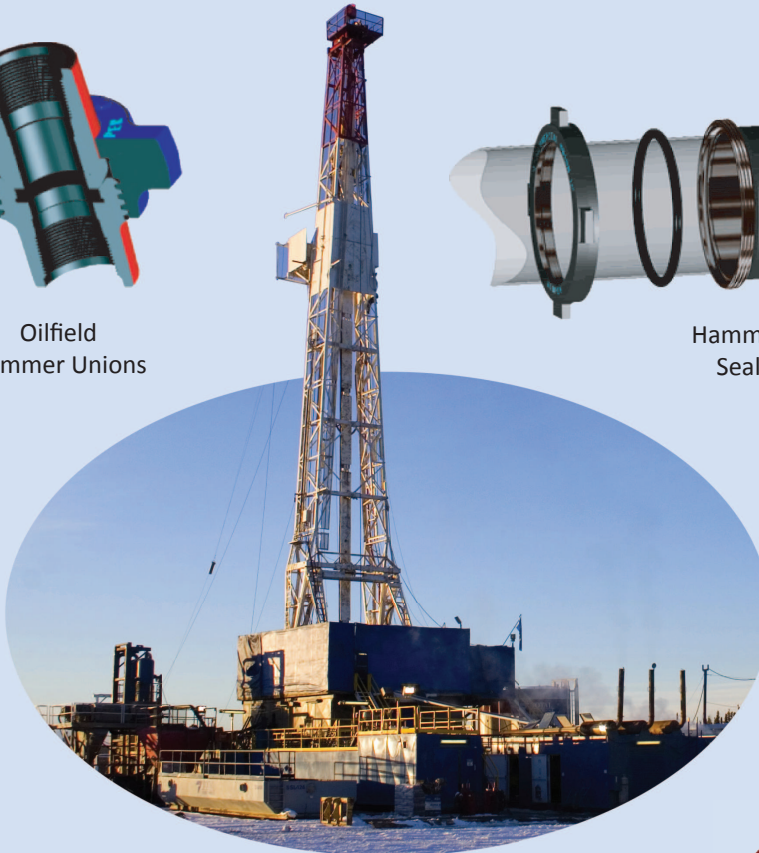
Oilfield  
Hammer Unions



Hammer  
Seal



Industrial  
Unions



Swivel Joints



Non-Freeze Tank Valves



Seal-O-Grip

## OILFIELD SERVICE AND DISTRIBUTION PRODUCTS

**Kemper Valve & Fittings Corporation** is a rapidly growing and leading provider of oilfield valves, hammer unions, swivel joints, oilfield flowline components, manifolds and complete rebuild / recertification services. This brochure outlines the products that are distributed through leading oilfield supply stores in North America. See our companion brochure "Flow Control and Stimulation Products" for additional products that are specifically designed and sold to the well servicing segment of the oil industry.



ISO 9001 Registered  
DNV Registered



## INDEX

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| Swivel Joints - Code Silver ..... | Page 7     |
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| Non-Freeze Tank Valves .....      | Page 8-9   |
| Industrial Unions .....           | Page 10-11 |

### Sales & Distribution

12343 Cutten Road  
**Houston, TX 77066**  
Local: 281-781-2730  
Toll Free: 800-231-2724  
Fax: 281-781-2731

### Sales & Service

2733 Industrial Drive  
**Sulphur, OK 73086**  
Local: 580-247-9623  
Toll Free: 888-533-3722  
Fax: 580-622-2058

### Sales & Service

1423 4th Avenue  
**Grand Junction, CO 81501**  
Local: 970-609-5845  
Fax: 970-609-5839

### Manufacturing HQ

3001 Darrell Road  
**Island Lake, IL 60042**  
Local: 847-526-2166  
Toll Free: 800-323-1659  
Fax: 847-526-2241

### Sales & Service

111 Arrowhead Drive  
Unit C  
**Slippery Rock, PA 16057**  
Local: 724-738-0130  
Fax: 724-738-0172

### Sales & Service

1000 West 2nd Street  
**Odessa, TX 79763**  
Local: 432-301-9990  
Fax: 432-301-9991

### Sales & Service

180 Sherman Way  
**Kensett (Searcy), AR 72082**  
Local: 501-268-2698  
Fax: 501-268-2196

### Sales & Distribution

1243 East Highway 40  
Unit 1  
**Vernal, UT 84078**  
Local: 435-781-8258  
Fax: 435-781-3037

### Canadian Sales

8001 Edgar Industrial Avenue  
**Red Deer, Alberta T4P 3S2**  
Local: 403-348-0872  
Fax: 403-348-0873

### Sales & Service

1391 County Road 429  
**Pleasanton, TX 78064**  
Local: 210-446-5863  
Fax: 830-569-8236

## OILFIELD UNIONS

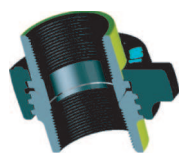


Figure 100

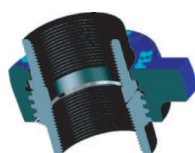


Figure 200

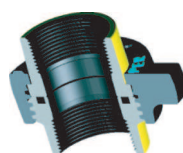


Figure 300

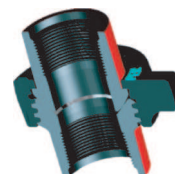


Figure 400

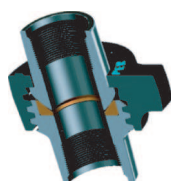


Figure 600

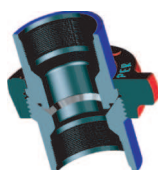


Figure 1002

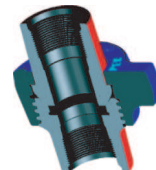


Figure 1502

| Figure Number    | Standard Service Color Code (Subs - Nut)                                            |                                                                                     | Non-Shock Cold Working Pressure Rating (psi) |       |             |       | Nominal Pipe Size (Inches)                                                          |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |
|------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------|-------|-------------|-------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|                  |                                                                                     |                                                                                     | Standard Service                             |       | H2S Service |       | 1                                                                                   | 1-1/4                                                                               | 1-1/2                                                                               | 2                                                                                   | 2-1/2                                                                               | 3                                                                                   | 4                                                                                   | 5                                                                                   | 6                                                                                   | 8                                                                                   | 10                                                                                  | 12                                                                                  |
|                  |                                                                                     |                                                                                     | CWP                                          | Test  | CWP         | Test  |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |
| 100 <sup>1</sup> |    |    | 1000                                         | 1500  |             |       |                                                                                     |                                                                                     |                                                                                     |                                                                                     |    |    |    |    |    |    |                                                                                     |                                                                                     |
| 101              |    |    | 1000                                         | 1500  |             |       |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |    |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |
| 200              |   |   | 2000                                         | 3000  |             |       |    |    |    |    |    |    |    |    |    |    |    |                                                                                     |
| 201              |  |  | 2000                                         | 3000  |             |       |                                                                                     |                                                                                     |                                                                                     |  | 2                                                                                   |  |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |
| 206              |  |  | 2000                                         | 3000  |             |       |  |  |  |  |  |  |  |  |  |  |  |                                                                                     |
| 207              |  |  | 2000                                         | 3000  |             |       |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |  |  |                                                                                     |  |                                                                                     |                                                                                     |                                                                                     |
| 211              |  |  | 2000                                         | 3000  |             |       |                                                                                     |                                                                                     |                                                                                     |  |                                                                                     |  |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |
| 300              |  |  | 2000                                         | 3000  |             |       |                                                                                     |                                                                                     |                                                                                     |  |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |
| 400              |  |  | 4000                                         | 6000  | 4000        | 6000  |                                                                                     |                                                                                     |                                                                                     |  |                                                                                     | 2                                                                                   | 2                                                                                   |  |  |                                                                                     |                                                                                     |  |
| 402              |  |  | 4000                                         | 6000  | 4000        | 6000  |                                                                                     |                                                                                     |                                                                                     |  |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |
| 600              |  |  | 6000                                         | 9000  |             |       |  |  |  |  |                                                                                     |                                                                                     |  |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |
| Figure G         |  |  | 6000                                         | 9000  | 6000        | 9000  |                                                                                     |                                                                                     |  |  |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |
| 602              |  |  | 6000                                         | 9000  | 6000        | 9000  |  |  |  |  |                                                                                     |  | 2                                                                                   |  |  |                                                                                     |                                                                                     |                                                                                     |
| 1002             |  |  | 10000                                        | 15000 | 7500        | 12000 |  |  |  |  |                                                                                     |  |  |  |  |                                                                                     |                                                                                     |                                                                                     |
| 1003             |  |  | 7500                                         | 12000 | 5000        | 7500  |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |  |  |  |  |                                                                                     |                                                                                     |                                                                                     |
| 1502             |  |  | 15000                                        | 22500 | 10000       | 15000 |  |                                                                                     |  |  |                                                                                     |  |  |  |                                                                                     |                                                                                     |                                                                                     |                                                                                     |
| 2002             |  |  | 20000                                        | 30000 |             |       |                                                                                     |                                                                                     |                                                                                     |  |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |
| 2202             |  |  |                                              |       | 15000       | 22500 |                                                                                     |                                                                                     |                                                                                     |  |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |

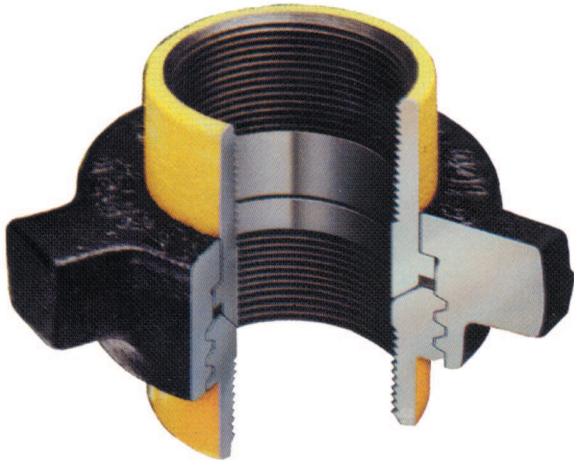
1 - ALSO AVAILABLE IN 16" FIGURE 100 w/ BUTTWELD SCHEDULE 80 ENDS

2 - ALSO AVAILABLE IN API 8 OR 10 ROUND THREADED ENDS

 3 - MINIMUM ORDER QUANTITIES MAY BE REQUIRED FOR UNIONS PREPPED FOR SOUR GAS SERVICE, OR OTHER NON-STANDARD UNION CONFIGURATIONS  
 PLEASE CONSULT MANUFACTURING, DISTRIBUTION OR STIMULATION SALES FOR AVAILABILITY, MINIMUM ORDER QUANTITIES, AND PRICING

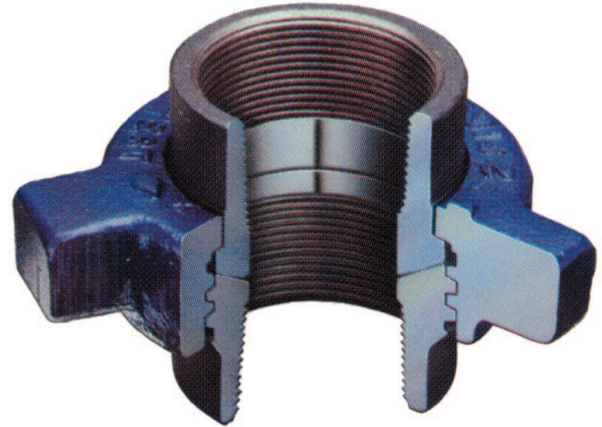
## **OILFIELD HAMMER UNIONS**

**FIGURE 100**  
**BLACK NUT, YELLOW SUBS**



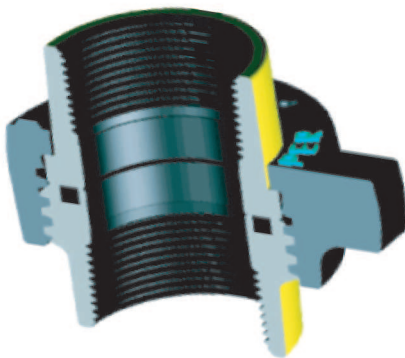
**Figure 100.** 1,000-psi NSCWP unions are available in 2-inch through 8-inch. Recommended for low-pressure manifolds and applications where working pressures do not exceed 1,000-psi. Figure 100 Unions are made from tough, impact resistant ductile iron, with precisely manufactured ball-cone (male-to-female) seats. This union is also available as a 16" butt-weld Schedule 80.

**FIGURE 200**  
**BLUE NUT, GRAY SUBS**



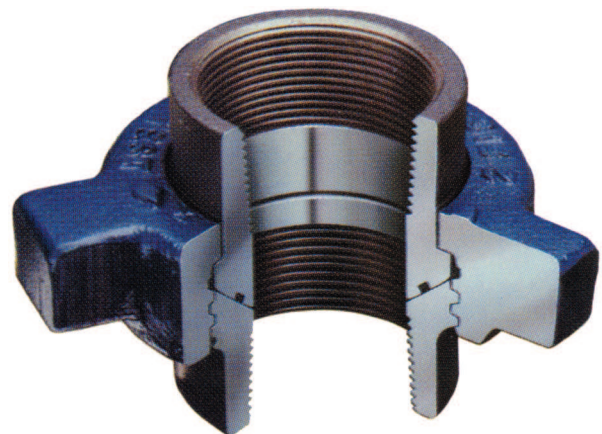
**Figure 200.** 2,000-psi NSCWP unions are available in 1-inch through 10-inch. These unions are compact and popular for low pressure rig piping. They also meet most requirements in the medium pressure ranges for air, water, oil or gas service. Available in butt-weld Schedule 40 and 80. Also available with stainless steel subs.

**FIGURE 300**  
**GRAY NUT, YELLOW SUBS**



**FIGURE 300.** 2,000-psi NSCWP unions are available in 2-inch line pipe thread size and are also available in 2" and 2½" 8 round EUE thread sizes. These unions feature the effective and dependable "flat-face/o-ring" sealing design. They seal equally well in low or high pressure vapor or liquid service. The "flat-face" design also permits straight away lateral breakouts of short rigid piping sections, without the need to spread the pipe.

**FIGURE 206**  
**BLUE NUT, GRAY SUBS**

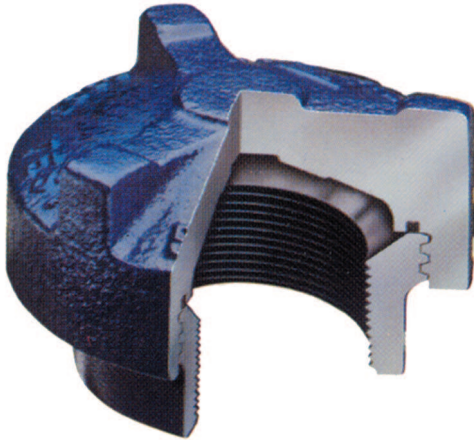


**Figure 206.** 2,000-psi NSCWP unions are available in 1-inch through 10-inch. These unions are fitted with an o-ring in the male sub seating surface for enhanced sealing. Available in butt-weld Schedule 40 and 80. Also available with stainless steel subs.



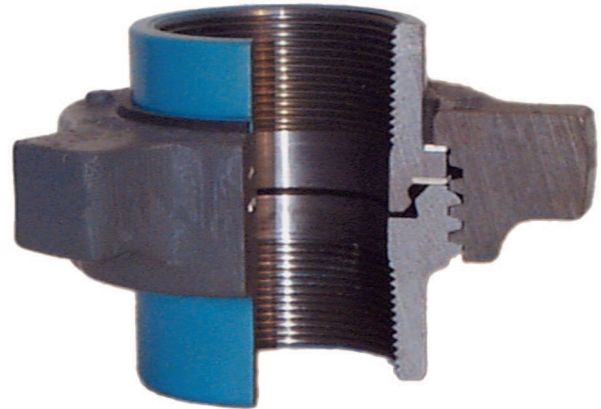
## OILFIELD HAMMER UNIONS

**FIGURE 207**  
**BLUE CAP, GRAY SUB**



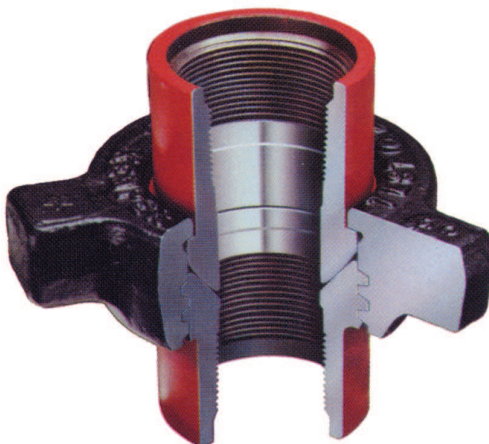
**Figure 207.** 2,000-psi NSCWP unions are available in 3, 4 and 6-inch. These unions are fitted with a nitrile o-ring in a blanking cap for effective closure. With the cap removed, the female sub will make up to hose, pipe or tubing fitted with a Figure 200 or 206 male sub and nut. Recommended for air, water, oil or gas service. Available in butt-weld Schedule 40 and 80.

**FIGURE 211**  
**GRAY NUT, BLUE SUBS**



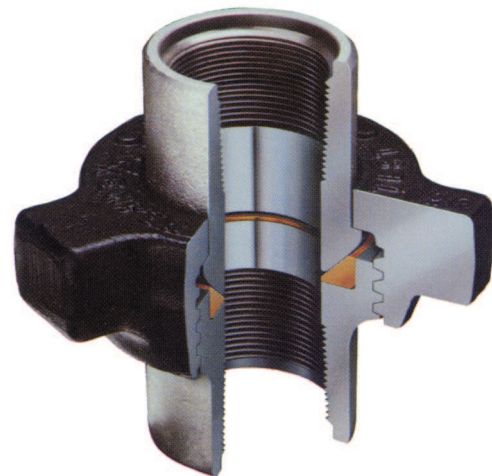
**Figure 211.** 2,000-psi NSCWP Insulating Unions are available in 2 and 3-inch sizes. The insulated design eliminates metal to metal contact across the union, while dual abrasion resistant precision molded elastomers provide a positive seal. Recommended where electrolytic corrosion prevention is a problem, or to statically isolate sections of piping. Replacement seal kits are available to extend service of the union.

**FIGURE 400**  
**BLACK NUT, RED SUBS**



**Figure 400.** These unions are available in 2, 3, 4, 5 and 6-inch 4,000-psi and 12-inch 2,500-psi NSCWP. All three parts are made from steel forgings with heavy walls and a rugged design. They feature ball and cone sealing for easy alignment and dependable seals. Also available in butt-weld Schedule 80 and 3" or 4" API 8 or 10 round threaded ends.

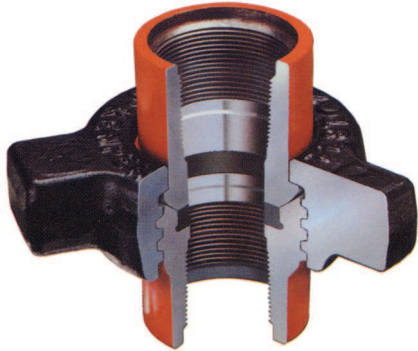
**FIGURE 600**  
**BLACK NUT, SILVER SUBS**



**Figure 600.** 6,000-psi NSCWP unions are available in 1, 1 ¼, 1 ½, 2, and 4-inch sizes. These unions have a bronze seat insert to provide an additional hardness differential across the seats for enhanced sealing.

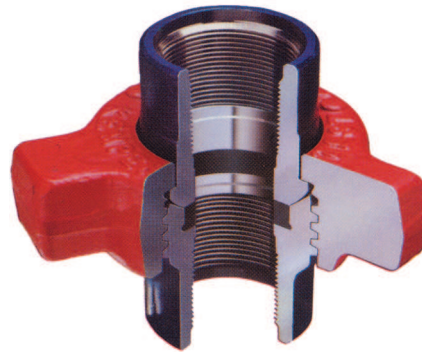
## OILFIELD HAMMER UNIONS

**FIGURE 602**  
**BLACK NUT, ORANGE SUBS**



**Figure 602.** 6,000-psi NSCWP unions are available in 1, 1½, 1¾, 2, 3, 4 and 5-inch sizes in standard or sour gas service. These unions feature a resilient nitrile seal ring that helps sealing and protects the steel-to-steel seating of the union. Recommended for manifold and truck-mounting where size and weight are critical. Fig. 602 unions are also available as non-pressure seal unions and in butt-weld Sch. 80, Sch. 160 and XXH, 4" API 8 round threaded ends.

**FIGURE 1002**  
**RED NUT, BLUE SUBS**



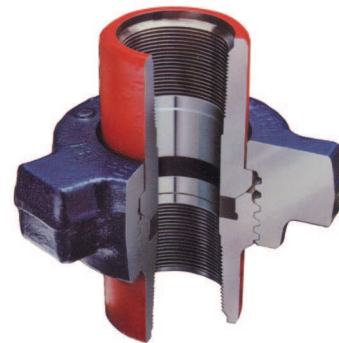
**Figure 1002.** These unions are available in 1 thru 4-inch 10,000-psi and 5 and 6-inch 7,500-psi NSCWP. These unions also have a resilient nitrite seal ring (5-inch and 6-inch have nitrile o-ring). They are made from alloy steel and are used primarily by service companies in applications such as cementing, fracturing and acidizing. Designed for high-pressure systems, including truck-mounted systems. Fig. 1002 unions also are available as non-pressure seal unions and in butt-weld Sch. 160, XXH or prepped for sour gas service.

**FIGURE 1003**  
**BLACK NUT, GREEN SUBS**



**Figure 1003.** Available in 3-inch rated at 10,000-psi and 4, 5 and 6-inch rated at 7,500-psi CWP. 3 and 4-inch are available in LPT threaded ends and all sizes in butt-weld Sch. 160 or XXH. Recommended for air, water, oil, mud or gas service where alignment of piping is a problem. Especially suitable for high pressure manifolding and drilling rig use. These unions incorporate a ball seat that provides for 7½° of misalignment or angular adjustment from center, for a total of 15° misalignment capability. In addition to the steel-to-steel seating of the union, a nitrile o-ring assures a pressure tight connection in any position of misalignment.

**FIGURE 1502**  
**BLUE NUT, RED SUBS**



**Figure 1502.** These 15,000-psi NSCWP unions are available in 1, 1½, 2, 3, 4 and 5-inch sizes with a replaceable resilient nitrite seal ring. Alloy steel construction, heavy walls and a rugged design have made these unions popular with the service companies for working pressures up to 15,000-psi. Figure 1502 unions also are available with non-pressure seal threads and in butt-weld Schedule 160 or XXH, 2" or 2½" API 8 round threaded ends or prepped for sour gas service.

*All Kemper o-ring and seal rings are available in standard NBR material or upon request, Viton or Highly Saturated Nitrile.*

## SWIVEL JOINTS - 6000 CWP

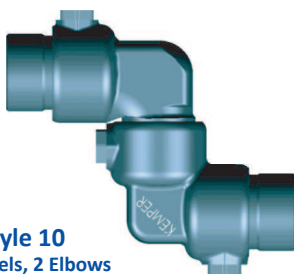
### CODE SILVER, STYLE 50



### CODE SILVER, STYLE 70



- High Pressure Swivel Joints - 6,000-psi CWP (Cold Working Pressure) in ¾", 1", 1¼", 1½" and 2" LPT sizes.
- For test lines, fracturing, acidizing, cementing and well servicing up to 6,000-psi CWP.
- Swivel Joints can be custom configured in any style for quick delivery.
- Manufactured to the highest quality standards in the industry.
- All supporting documentation, including MTRs, is available at any time for *no additional cost*.
- Maintenance, rebuild and recertification are available through our Service facilities.



**Style 10**  
3 Swivels, 2 Elbows



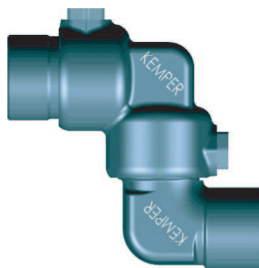
**Style 20**  
1 Swivel



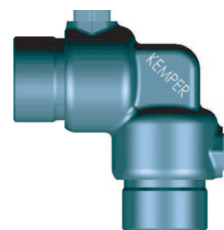
**Style 30**  
1 Swivel, 1 Elbow



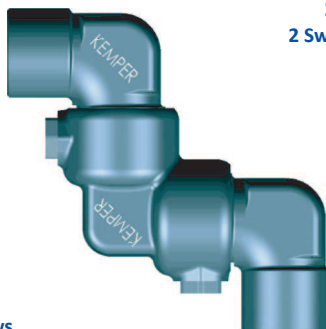
**Style 40**  
1 Swivel, 2 Elbows



**Style 50**  
2 Swivels, 2 Elbows



**Style 60**  
2 Swivels, 1 Elbow



**Style 70**  
2 Swivels, 3 Elbows

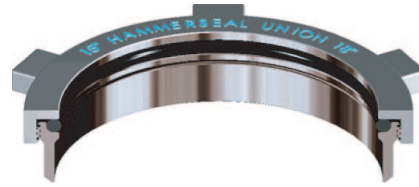


**Style 80**  
3 Swivels, 3 Elbows

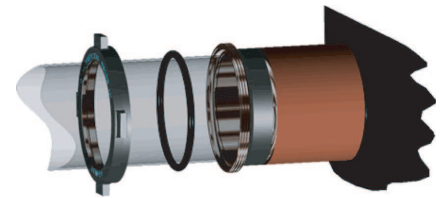


## **Hammerseal™ Unions** Join drilling mud tanks in a hurry

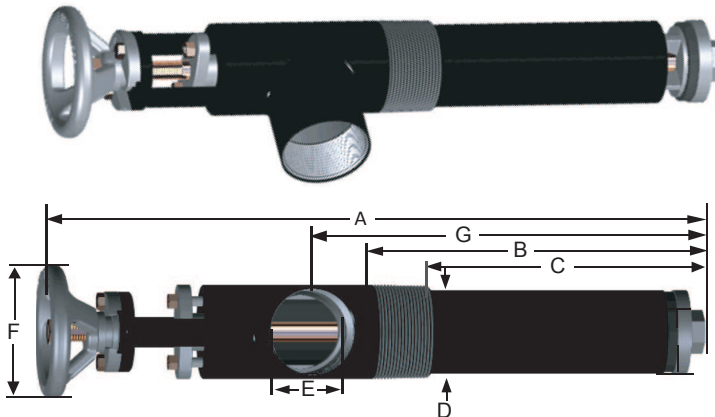
Even if mud tanks aren't aligned perfectly (up to 7°), you can hammer these tough unions tight to a non-leak seal. The externally threaded female portion of the HammerSeal™ welds to a Schedule 80 stub pipe at the tank wall. The connecting pipe (nominal size defines the union) extends through the nut component, the o-ring and then the female portion of the union. When the internally threaded lug nut is hammered tight onto the female, the o-ring is compressed into an oblong cross section to seal up to 150-psi against the outside diameter of the connecting pipe.



3" and 20" HammerSeals™ are Available -  
Please Consult Factory



## **Non-Freeze Tank Valves** Give trouble-free service in cold country



The only precision-manufactured non-freeze storage tank valves in 2-inch and 3-inch sizes (4-inch valve available on special order only). These valves operate in extremely cold climates when other tank valves freeze up. The stainless steel valve stem meets most corrosive service requirements. Adjustable gland prevents leaking and extends the wear-life of the gland packing. A heavy duty nitrile valve seat and nitrile gland packing tolerate hydrocarbon liquids for long periods without deterioration. Standard line pipe connections permit easy make-up at the tank connection and valve outlet. Operating pressure is up to 150-psi (10.5 kg/cm<sup>2</sup>). High torque handwheel makes opening and closing easy at all times. Complete valves and individual parts available. All sizes also available with flange connections.

## **Seal-O-Grip™ Connections** Provide air-inflatable tight seal between mud tanks



Seal-O-Grip™ Unions are used for speedy connect-and-disconnect of mud suction and return lines or other low-pressure fluid lines. A tight seal is maintained even if the pipe is mis-aligned, within certain limits. Available in popular oilfield line pipe sizes: 4, 6, 8, 10, 12 and 16-inch.

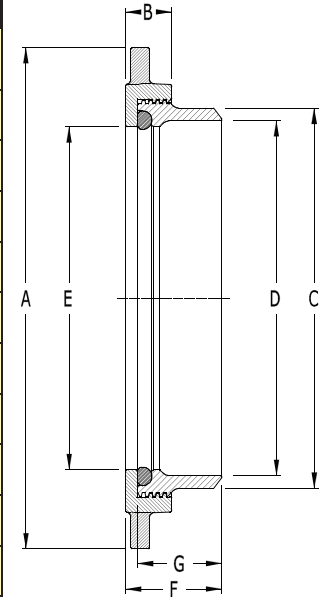
The Seal-O-Grip™ body (ASTM A216 Grade WCB carbon steel) welds in place on mud tanks, manifold pipe and other fittings. The oil-resistant inflatable tube is heavy duty, abrasion-resistant and is designed for 200-psi maximum air pressure. (Note: Tube inflation pressure must exceed fluid line pressure by 50-psi to seal line pressures up to 150-psi.)

*Note: See tables on next page for HammerSeal™ and Seal-O-Grip™ dimensional information.*



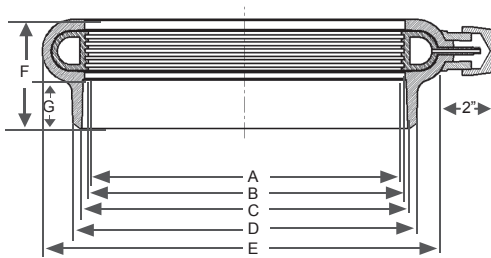
## Hammerseal™ Union Dimensions

| Connects<br>Nominal Size<br>Oilfield Pipe: | Female<br>Component<br>Welds To: | Complete Assembly    |                          | O-Ring Only   |                          | Dimensions in Inches |      |       |       |       |      |      |
|--------------------------------------------|----------------------------------|----------------------|--------------------------|---------------|--------------------------|----------------------|------|-------|-------|-------|------|------|
|                                            |                                  | Assem-<br>bly<br>P/N | Approx<br>Weight<br>lbs. | O-Ring<br>P/N | Approx<br>Weight<br>lbs. | A                    | B    | C     | D     | E     | F    | G    |
| 3-Inch<br>(3 1/2" O.D.)                    | 4-Inch                           | PS300A               | 15                       | PS300R        | 0.38                     | 9.75                 | 2    | 4.5   | 3.94  | 3.75  | 3.5  | 3.0  |
| 4-Inch<br>(4 1/2" O.D.)                    | 6-Inch                           | PS400A               | 22                       | PS400R        | 0.5                      | 10.75                | 2    | 6.63  | 5.88  | 4.81  | 3.5  | 3.0  |
| 6-Inch<br>(6 5/8" O.D.)                    | 8-Inch                           | PS600A               | 37                       | PS600R        | 0.75                     | 13.5                 | 2    | 8.63  | 7.88  | 6.88  | 5.0  | 4.5  |
| 8-Inch<br>(8 5/8" O.D.)                    | 10-Inch                          | PS800A               | 42                       | PS800R        | 1.0                      | 15.75                | 2    | 10.75 | 10    | 8.88  | 5.0  | 4.5  |
| 10-Inch<br>(10 3/4" O.D.)                  | 12-Inch                          | PS100A               | 53                       | PS100R        | 1.25                     | 17.75                | 2    | 12.75 | 12    | 11    | 5.0  | 4.5  |
| 12-Inch<br>(12 3/4" O.D.)                  | 14-Inch                          | PS120A               | 61                       | PS120R        | 1.5                      | 19.75                | 2    | 14    | 13.25 | 13    | 5.0  | 4.5  |
| 13-Inch<br>(13 3/8" O.D.)                  | 13-Inch<br>(Casing)              | PS130A               | 53                       | PS130OR       | 1.6                      | 19.75                | 2    | 14.13 | 13.5  | 13.5  | 5.0  | 4.5  |
| 14-Inch<br>(14" O.D.)                      | 16-Inch                          | PS140A               | 83                       | PS140R        | 1.75                     | 22.0                 | 2    | 16    | 15    | 14.25 | 4.5  | 4.0  |
| 16-Inch<br>(16" O.D.)                      | 18-Inch                          | PS160A               | 99                       | PS160R        | 2.0                      | 24.0                 | 2.25 | 18    | 16.5  | 16.25 | 4.56 | 4.0  |
| 18-Inch<br>(18" O.D.)                      | 20-Inch                          | PS180A               | 110                      | PS180R        | 2.25                     | 26.0                 | 2.25 | 20    | 18.8  | 18.25 | 4.56 | 4.0  |
| 20-Inch<br>(20" O.D.)                      | 22-Inch                          | PS200A               | 140                      | PS200R        | 3                        | 28.0                 | 2.25 | 22    | 20.8  | 20.25 | 5.0  | 5.63 |



## Non-Freeze Tank Valve Dimensions

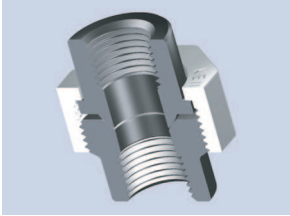
| Complete Valve | Non-Freeze Valve Dimensions in Inches |    |        |       |   |       |   |        |       |       |
|----------------|---------------------------------------|----|--------|-------|---|-------|---|--------|-------|-------|
| Part No.       | Valve Size                            | A  | B      | C     | D | E     | F | G      | H     | I     |
| N-F200         | 2                                     | 22 | 10 5/8 | 9 1/4 | 2 | 2 1/8 | 5 | 13 1/4 | 2 1/8 | 2 3/8 |
| N-F300         | 3                                     | 25 | 12     | 10    | 3 | 3 1/8 | 5 | 15 1/4 | 3 1/8 | 3 1/2 |



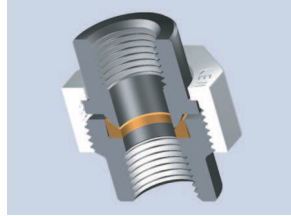
## Seal-O-Grip™ Connector Dimensions

| Line Pipe<br>Size<br>(inches) | Seal-O-Grip<br>Assembly Complete |                     | Inflatable<br>Tube Only |                     | Maximum<br>Mis-Alignment<br>Tolerance | Seal-O-Grip Dimensions (Inches) |         |        |        |        |       |       |
|-------------------------------|----------------------------------|---------------------|-------------------------|---------------------|---------------------------------------|---------------------------------|---------|--------|--------|--------|-------|-------|
|                               | Part No.                         | Approx.<br>Wt (lbs) | Part No.                | Approx.<br>Wt (lbs) |                                       | A                               | B       | C      | D      | E      | F     | G     |
| 4                             | P480                             | 5.3                 | P480T                   | .8                  | 7                                     | 4 19/32                         | 4 19/32 | 5 1/6  | 5 3/4  | 6 5/8  | 3     | 1 1/2 |
| 6                             | P680                             | 23.0                | P680T                   | 1.4                 | 14                                    | 6 3/4                           | 6 15/16 | 8      | 8 5/8  | 10 1/4 | 4     | 2     |
| 8                             | P880                             | 27.0                | P880T                   | 1.6                 | 14                                    | 8 3/4                           | 8 15/16 | 10 1/8 | 10 3/4 | 12 1/4 | 4     | 2     |
| 10                            | P1080                            | 31.0                | P1080T                  | 2.4                 | 14                                    | 10 15/16                        | 11 1/8  | 12 1/8 | 12 3/4 | 14 1/2 | 4 1/4 | 2     |
| 12                            | P1280                            | 38.5                | P1280T                  | 3.2                 | 14                                    | 12 15/16                        | 13 1/8  | 13 3/8 | 14 1/4 | 16 1/2 | 4 1/4 | 2     |
| 14                            | P1480                            | 42.0                | P1480T                  | 3.3                 | 14                                    | 14 1/8                          | 14 1/4  | 15 1/4 | 16 1/4 | 18 1/4 | 4 1/2 | 2     |
| 16                            | P1680                            | 46.5                | P1680T                  | 3.5                 | 14                                    | 16 1/8                          | 16 1/4  | 17 1/4 | 18 1/4 | 20 1/4 | 5 1/8 | 2     |

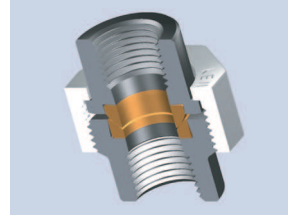
## INDUSTRIAL UNIONS - ½" thru 3" Nominal Pipe Sizes



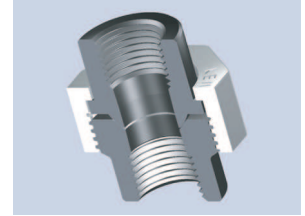
**3000# "AAR" Unions**  
**Steel-to-Steel**  
Threaded or Socket Weld



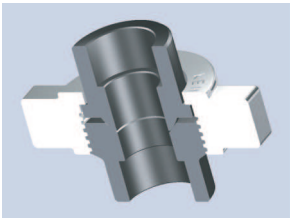
**3000# Unions**  
**Bronze-to-Steel**  
Threaded or Socket Weld



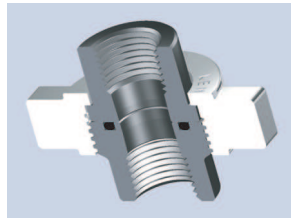
**3000# Unions**  
**Bronze-to-Bronze**  
Threaded or Socket Weld



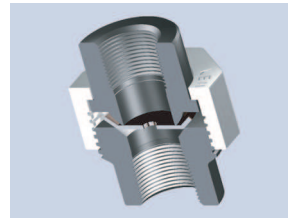
**3000# Hi-Speed Unions**  
**Steel-to-Steel**  
Threaded or Socket Weld



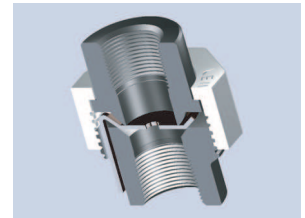
**3000# Hi-Speed Hammer-Lug**  
**Nut Unions**  
**Steel-to-Steel**  
Threaded or Socket Weld



**3000# Hi-Speed Hammer-Lug**  
**Nut Unions**  
**Flat-Face to O-Ring**  
Threaded or Socket Weld

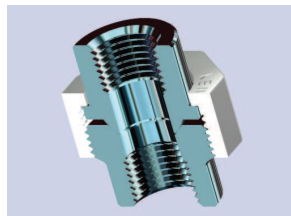


**3000# Hi-Speed Orifice Plate**  
**Unions**  
**Steel-to-Plate-to-Steel**  
Threaded or Socket Weld

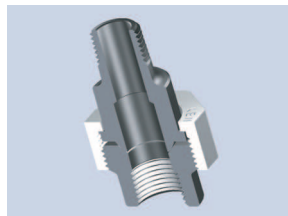


**3000# Hi-Speed "Tabbed"**  
**Orifice Plate Unions**  
**Steel-to-Plate-to-Steel**  
Threaded or Socket Weld

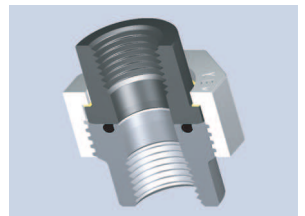
*ASTM A182 3000# "AAR" Style Unions in Austenitic Stainless Steel Grades 316, 316L, 304, 304L are available. Other A182 grades can be produced for special requirements. Please consult factory directly for pricing and delivery.*



**3000# "AAR" Unions**  
**ASTM A182 Stainless**  
**Steel-to-Steel**  
Threaded or Socket Weld



**3000# "AAR" Unions**  
**Steel-to-Steel**  
Threaded Male X Female

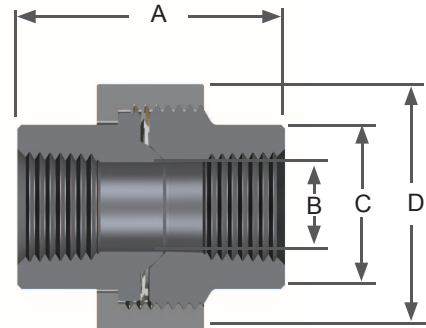


**3000# Insulated Unions Flat-**  
**Face to O-Ring**  
Threaded

***Kemper 3000# Forged Steel Industrial Unions  
conform to one or more of the following specifications:***

- ASTM A105 "Forgings, Carbon Steel, for Piping Components"
- ASTM A182 "Forged or Rolled Alloy-Steel Pipe Flanges, Forged Fittings, and Valve and Parts for High-Temperature Service"
- MSS SP-83 "Carbon Steel Pipe Unions, Socket Welding and Threads"
- MSS SP-25 "Standard Marking Systems for Valves, Fittings, Flanges and Unions"
  - ANSI/ASME B1.20.0 "Pipe Threads, General Purpose"
  - ANSI/ASME B1.8 "Stub Acme Screw Threads"
  - ANSI/ASME B1.1 "Unified Inch Screw Threads"
- NACE MR01-75 "Sulfide Stress Cracking Resistant Metallic Materials for Oilfield Equipment"

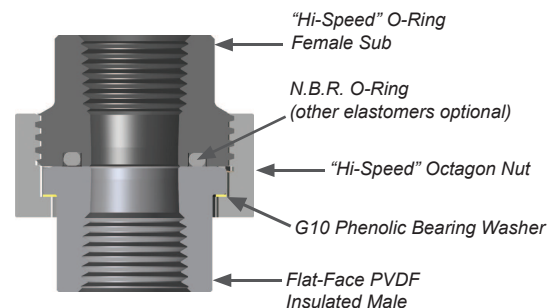
| MSS SP-83 Dimensions and Weights |      |      |      |      |          |
|----------------------------------|------|------|------|------|----------|
| Size                             | A    | B    | C    | D    | lb / 100 |
| 1/4                              | 1.65 | .380 | .78  | 1.38 | 34       |
| 3/8                              | 2.00 | .540 | 1.09 | 1.75 | 72       |
| 1/2                              | 2.13 | .680 | 1.25 | 1.81 | 69       |
| 3/4                              | 2.35 | .850 | 1.50 | 2.20 | 117      |
| 1                                | 2.60 | 1.10 | 1.81 | 2.56 | 167      |
| 1 1/4                            | 3.00 | 1.38 | 2.17 | 3.06 | 250      |
| 1 1/2                            | 3.19 | 1.63 | 2.44 | 3.42 | 325      |
| 2                                | 3.53 | 2.06 | 2.97 | 4.09 | 500      |
| 2 1/2                            | 4.13 | 2.54 | 3.63 | 4.88 | 815      |
| 3                                | 4.50 | 3.05 | 4.31 | 5.75 | 1175     |



***Kemper "Insulated" Flat-Face O-Ring Industrial Unions***

Kemper Insulated Unions provide galvanic corrosion protection for piping systems. A tough Polyvinylidene Fluoride polymer is chemically bonded to the flat-faced male component and a phenolic bearing washer sits between the male flange and nut shoulder. PVDF is a high molecular weight thermoplastic polymer with excellent chemical inertness, high dielectric strength and thermal stability.

| Properties of Kemper Industrial Union Insulation |                            |
|--------------------------------------------------|----------------------------|
| Property                                         | Capability                 |
| Temperature                                      | -20° F to 300° F           |
| Dielectric constant at 60Hz                      | (73° F, 50% RH): 9.0       |
| Dissipation factor at 60Hz                       | (73° F) : 0.06             |
| Volume resistivity                               | 5.0x10~14 ohm-cm           |
| Dielectric strength                              | 280 v/MIL                  |
| Water & Oil Absorption (%)                       | <0.04%                     |
| Thermal conductivity                             | 1.32 Btu-inch/hr-ft~2- ° F |
| Coefficient of linear thermal expansion          | 7.1x10 <sup>-6</sup> Max.  |
| Deformation under load                           | 0.5% @ 4,000 psi           |





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